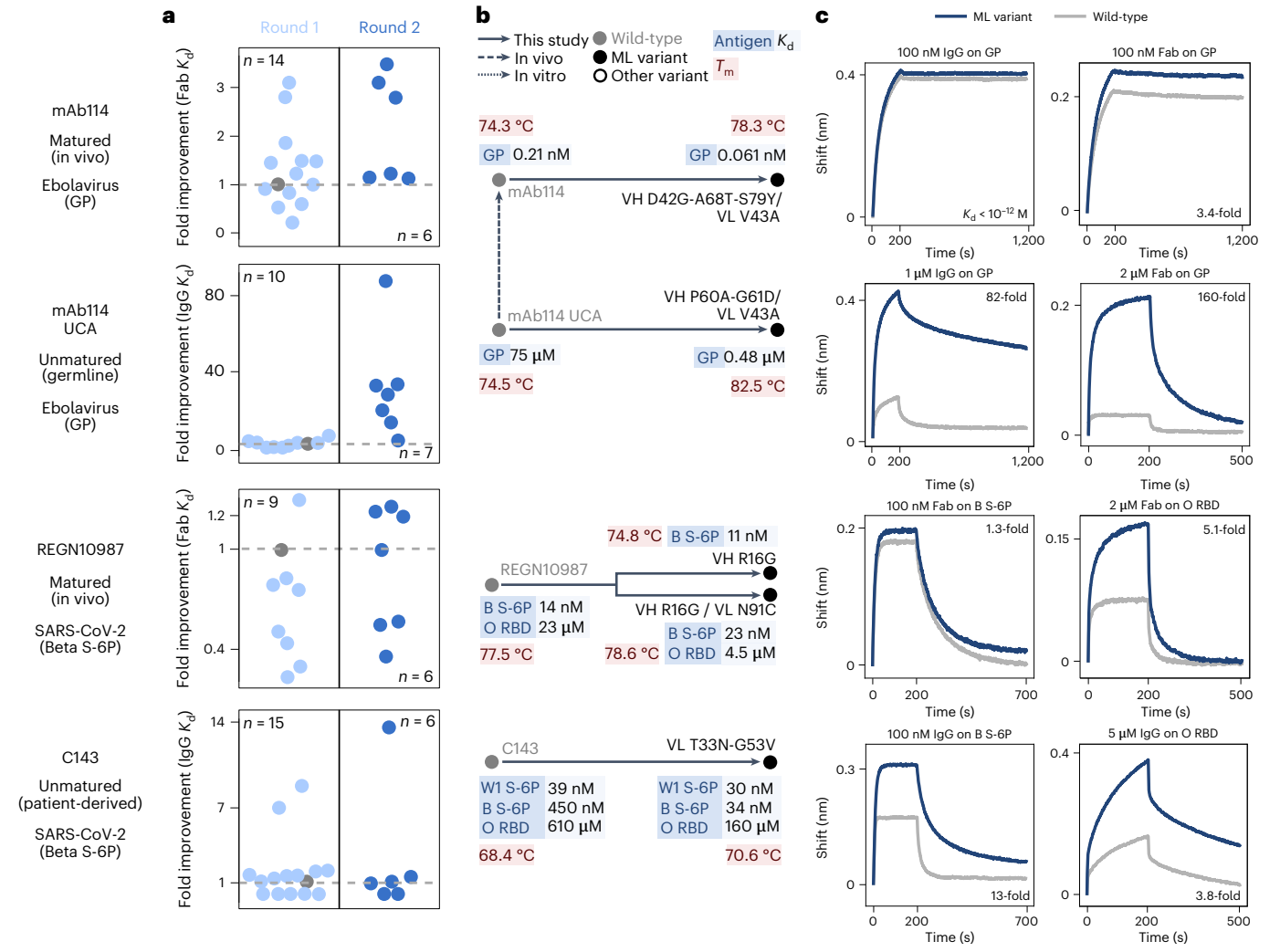
a
Round 1
Round 2
mAb114
Matured (in vivo)
Ebolavirus (GP)
Fold improvement (Fab K_d)
n = 14
n = 6
mAb114 UCA
Unmatured (germline)
Ebolavirus (GP)
Fold improvement (IgG K_d)
n = 10
n = 7
REGN10987
Matured (in vivo)
SARS-CoV-2 (Beta S-6P)
Fold improvement (Fab K_d)
n = 9
n = 6
C143
Unmatured (patient-derived)
SARS-CoV-2 (Beta S-6P)
Fold improvement (IgG K_d)
n = 15
n = 6
b
This study
In vivo
In vitro
Wild-type
ML variant
Other variant
Antigen K_d
T_m
74.3 °C
78.3 °C
GP 0.21 nM
GP 0.061 nM
mAb114
VH D42G-A68T-S79Y/ VL V43A
VH P60A-G61D/ VL V43A
mAb114 UCA
GP 75 μM
GP 0.48 μM
74.5 °C
82.5 °C
74.8 °C
B S-6P 11 nM
VH R16G
REGN10987
B S-6P 14 nM
O RBD 23 μM
VH R16G / VL N91C
B S-6P 23 nM
O RBD 4.5 μM
77.5 °C
78.6 °C
C143
VL T33N-G53V
W1 S-6P 39 nM
B S-6P 450 nM
O RBD 610 μM
W1 S-6P 30 nM
B S-6P 34 nM
O RBD 160 μM
68.4 °C
70.6 °C
c
ML variant
Wild-type
100 nM IgG on GP
$K_d < 10^{-12}$ M
100 nM Fab on GP
3.4-fold
1 μM IgG on GP
82-fold
2 μM Fab on GP
160-fold
100 nM Fab on B S-6P
1.3-fold
2 μM Fab on O RBD
5.1-fold
100 nM IgG on B S-6P
13-fold
5 μM IgG on O RBD
3.8-fold
Shift (nm)
Time (s)